

ATMA: Long-Context Language Modeling via Polar Attention and Gated-Delta Compression Memory

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Abstract

Length extrapolation in language models involves competing objectives: retrieval fidelity, long-document likelihood, short-context quality, and inference cost. We present ATMA, a 378M-parameter hybrid recipe that combines Polar Attention with gated-delta recurrent memory, and study these objectives as a Pareto problem rather than claiming general architectural dominance. Polar Attention separates a normalized direction channel from a bounded participation-ratio magnitude channel. We select the recipe with a complete 120-cell, 1B-token factorial sweep, then train matched NoPE, RoPE, and Polar variants for 9.816B tokens at length 2K and evaluate them through 256K. Across the factorial, memory improves Polar’s 64K retrieval score in all 20 matched cells (mean +47.8 points), whereas its effect on NoPE is small and inconsistent. At 256K, Polar retains 34.4% teacher-forced target-token accuracy and 9.0% exact five-token accuracy; exact retrieval is 18.0% on synthetic contexts but 0.0% on FinePDFs contexts. Polar also limits mean fixed-target bits-per-byte degradation to $1.26\times$, at a 1.9-point mean cost on eight short-context tasks. Raven baselines lead BABILong and have length-independent decode state, illustrating a different point on the frontier. Finally, a post-hoc checkpoint audit shows that nearly identical 2K validation curves can conceal a 6.70-nat difference at 256K. Because those runs were neither seed-paired nor randomized across devices, we interpret this as checkpoint variability associated with an infrastructure transition, not a causal hardware effect.

1 Introduction

Large language models increasingly process code repositories, books, legal documents, and scientific papers that exceed their pretraining sequence length. Full softmax attention preserves token-level access in principle, but its probability mass can disperse as the key set grows (Nakanishi, 2025; Velickovic et al., 2025; Barbero et al., 2024). Sliding windows bound cost by removing older tokens from direct access. Recurrent sequence models instead keep fixed-size state but compress history. Long-context modeling therefore exposes a multi-objective trade-off among retrieval fidelity, sequence likelihood, reasoning, short-context quality, and serving cost.

We present ATMA, a hybrid architecture that combines short gated convolutions with four Polar Attention layers as global mixers. Polar Attention represents “what matched” with a normalized direction and “how much matched” with a bounded participation-ratio channel. A learned null sink follows the extreme-value scale of irrelevant scores. Each global layer also contains gated-delta memory, providing a compressed recurrent history alongside full-context attention (Appendix A).

Our contributions are threefold. First, a complete 120-cell, 1B-token factorial sweep measures interactions among attention core, recurrent memory, training-time local windows, distractor alignment, and representation regularization. Second, we promote one recipe to matched 9.816B-token NoPE, RoPE, and Polar runs and evaluate retrieval, BABILong, fixed-target likelihood, short-context tasks, and serving. Raven models provide a strong but separately optimized recurrent-family comparison (Afzal et al., 2026). Figure 1 summa-

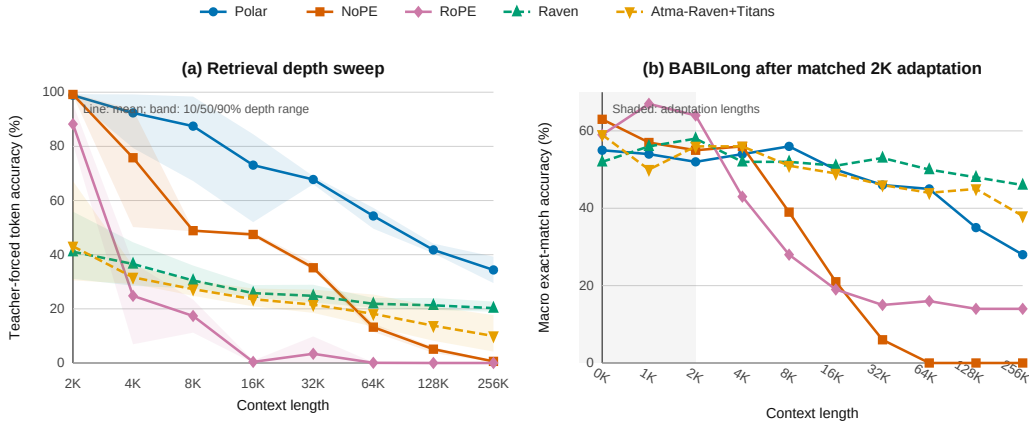


Figure 1: Results after 9.816B-token pretraining. Left: teacher-forced target-token retrieval accuracy, averaged over tasks, haystack suites, and needle depths. Right: BABILong exact match after adaptation. Polar retains more retrieval signal than the matched attention variants, whereas the separately optimized Raven family leads long-context BABILong.

rizes the resulting frontier: Polar improves long-context retrieval and likelihood among the matched attention variants, while Raven favors BABILong and decode efficiency. Third, a post-hoc audit demonstrates that short-context convergence alone did not predict extrapolation for three NoPE checkpoints. This audit motivates reporting extended-context behavior and execution metadata, but does not identify the device transition as causal.

2 Related Work

Softmax attention (Vaswani et al., 2017) has quadratic compute and can lose selectivity as the number of keys grows. RoPE (Su et al., 2021) and ALiBi (Press et al., 2021) alter positional information while retaining dense simplex normalization. Scalable-Softmax, adaptive-temperature analyses, and sparse alternatives address dispersion or length-dependent sharpness (Nakanishi, 2025; Velickovic et al., 2025; Vasylenko et al., 2026). Threshold Differential Attention (TDA) uses a $\sqrt{\log n}$ threshold motivated by maxima of noise scores (Huang et al., 2026). Polar combines length-dependent temperature and extreme-value scaling with a differentiable null sink and bounded magnitude channel. We compare Polar with matched NoPE and RoPE controls; comparisons with these other methods are conceptual rather than empirical.

Linear attention, state-space models, and gated recurrences exchange full-resolution token access for compressed fixed-size history (Gu & Dao, 2023; Yang et al., 2024b;a; Behrouz et al., 2025). Raven combines routed sparse memory, decay dynamics, and gated-delta recurrence (Afzal et al., 2026). We treat it as a strong sub-quadratic baseline, but not as an attention ablation because it uses a different model family and optimizer. Hybrid models such as LFM2 combine local and global mixers for efficiency (Hasani et al., 2025; Allen-Zhu, 2025). ATMA follows this hybrid pattern while changing the global mixer and adding a recurrent memory stream.

3 Method

3.1 Hybrid architecture

ATMA is a 16-layer decoder with 12 LFM2-style gated convolutional layers interleaved with four global attention layers. All blocks use pre-normalization and squared-ReLU MLPs. The attention variants share hidden size 1024, eight query heads, two key-value heads, head

dimension 128, tokenizer, data order, optimizer, schedule, and training budget; only the attention core changes among NoPE, RoPE, and Polar. Canon depthwise causal convolutions are applied to q , k , and v before attention (Allen-Zhu, 2025).

3.2 Polar attention

For query i and its $n_i = i + 1$ visible keys, let $\sigma_{ij} = q_i^\top k_j / \sqrt{d_k}$. Each head learns a length temperature and a virtual null logit,

$$t_i = 1 + \text{softplus}(\alpha) \log n_i, \quad \nu_i = b + \text{softplus}(\gamma) \sqrt{\log(n_i + 1)}. \quad (1)$$

The second term tracks the extreme-value scale of unrelated scores. Appending the null as a softmax competitor gives

$$w_i = \text{softmax}(t_i[\sigma_{i\bullet}, \nu_i]), \quad s_i = \sum_j w_{ij} v_j + w_{iN} v_{\text{null}}. \quad (2)$$

The direction channel is $c_i = s_i / \max(\|s_i\|_2, \varepsilon)$, hence $\|c_i\|_2 \leq 1$. To retain bounded information about match multiplicity, real-key weights are renormalized as $\hat{w}_{ij} = w_{ij} / \sum_k w_{ik}$ and summarized by the participation ratio

$$n_{\text{eff},i} = \left(\sum_j \hat{w}_{ij}^2 \right)^{-1}, \quad m_{\text{eff},i} = n_{\text{eff},i}(1 - w_{iN}), \quad \mu_i = \tanh(\text{softplus}(\beta) \log(1 + m_{\text{eff},i})). \quad (3)$$

Thus $\mu_i \in [0, 1)$, while the direction channel removes radial scale. This is a channel decomposition, not an independence claim: changing the match set can rotate c_i . The Polar output is a gated projection of c_i plus a learned projection of μ_i . These choices were selected through pre-integration falsification tests: additive soft counts accumulated leakage with length, fixed or relative floors were overtaken by noise tails, and an unbounded $\log m$ channel left the training range. The standalone prototype was not retained; Appendix C distinguishes this design provenance from the reproducible integrated oracle and kernel tests. A distractor-alignment auxiliary loss was tested in the factorial but is disabled in the promoted recipe.

3.3 Gated-delta memory

Each global layer also maintains a per-head recurrent state $M_t \in \mathbb{R}^{d_v \times d_k}$. With unit-normalized keys and queries, retention γ_t , and write gate β_t , the decay-first, self-inclusive recurrence is

$$M_t = \gamma_t M_{t-1} (I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top, \quad r_t = M_t q_t. \quad (4)$$

RMS-normalized r_t is gated, projected, and added to the Polar output. The projection is zero-initialized. Unit L_2 normalization is important: RMS-normalized keys have $\|k\|_2^2 = d_k$ and made the tested delta recurrence diverge, whereas unit keys keep the update eigenvalue along k within the gate-controlled range. Under bounded values and a uniformly contractive retention gate, Appendix A gives a sequence-length-independent state bound. Our empirical claim is narrower: the implemented state remains stable in the tested length sweep.

Figure 2 summarizes the two mechanisms and their controlled diagnostics. Appendix B defines every regularization mode in the factorial sweep, Appendix C gives the numerical oracle and streaming Triton implementation, and Appendix H details the kernel and serving optimizations.

4 Experimental Design

4.1 Recipe selection

Stage I trains all $3 \times 5 \times 2 \times 2 \times 2 = 120$ combinations of attention type (Polar, RoPE, NoPE), regularization, distractor loss, memory, and a 1024-token training window. Every model receives approximately 1B FineWeb-Edu tokens at length 2048 on an NVIDIA L4, and every evaluation uses full context. These are distinct factorial cells rather than seed replicates; Stage I selects the promoted recipe and is not used for headline absolute comparisons.

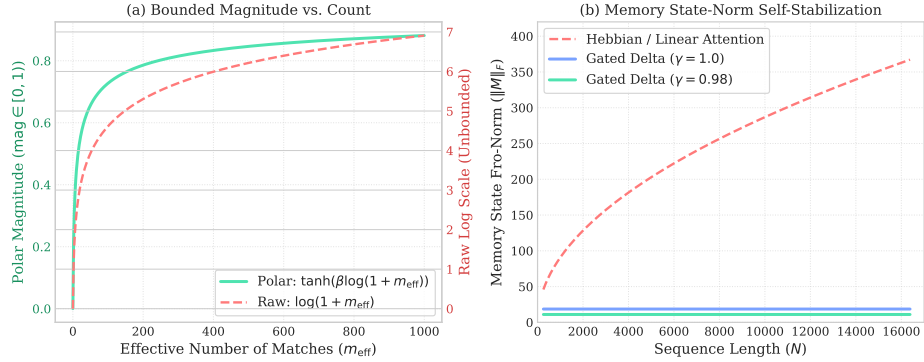


Figure 2: ATMA diagnostics. Left: the effective match count is mapped to a bounded magnitude channel. Right: the tested gated-delta state remains flat with length, unlike an ungated Hebbian memory. These controlled diagnostics illustrate the mechanism, they are not a network-level stability proof.

4.2 Matched comparison

Stage II trains 378.16–378.22M-parameter NoPE, RoPE, and Polar models for 9.816B tokens at length 2048 in the same L40S software and hardware environment. All three use the same Muon/AdamW parameter split. Raven Native (388.54M) and Atma-Raven-Titans (382.37M) use the same data, tokenizer, token budget, and device class, but use AdamW. This follows a 1B-token Raven Native pilot with the ATMA Muon schedule: validation loss improved to 3.76 nats at step 625, then destabilized and ended at 4.62, with 0% retrieval through 64K. We therefore report the stable Raven models as a separate model-family/optimizer group. The single failed pilot does not establish incompatibility with every Muon configuration.

Evaluation. The evaluation contains 24,000 paired retrieval trials over synthetic and FinePDFs haystacks, lengths 2K–256K, depths 10/50/90%, and 50 trials per cell; BABI-Long adaptation and exact match (Kuratov et al., 2024); fixed-target bits per byte (BPB) on FinePDFs, PG-19, and Proof-Pile; eight likelihood-scored downstream tasks (22,939 examples per model); and single-sequence prefill/decode on one L40S. Retrieval teacher-forces the five-token target and reports both token accuracy and exact five-token accuracy. Neither metric is autoregressive generation success. BABI-Long uses custom splits and prompts, so its absolute scores are internally comparable rather than leaderboard replications. Protocols, denominators, and complete matrices appear in Appendix D.

5 Results

5.1 Component selection

Table 1 summarizes matched contrasts from the full factorial rather than selecting individual cells. At 64K, adding memory improves Polar in all 20 combinations of regularizer, distractor setting, and window setting, by 47.8 points on average. The corresponding NoPE effect is small and heterogeneous, and RoPE remains at zero. Among memory-enabled cells, Polar is never worse than NoPE and exceeds RoPE in all 20 comparisons. A training window reduces Polar+memory by 25.5 points on average and is harmful in 8/10 matched cells; distractor alignment has a smaller, nonuniform effect (mean −6.0 points; harmful in 6/10). We therefore promote full-context Polar plus memory without distractor alignment. The selected baseline cell reaches 92.5% token accuracy at 64K; the complete matrix is in Appendix E.

64K matched contrast	Mean Δ	Median Δ	Range	Positive
Polar: memory on – off	+47.8	+51.3	[+7.5, +92.5]	20/20
NoPE: memory on – off	+4.6	+1.9	[−15.0, +21.3]	11/20
RoPE: memory on – off	0.0	0.0	[0.0, 0.0]	0/20
Polar+memory – NoPE+memory	+41.7	+41.9	[0.0, +77.5]	19/20 (+1 tie)
Polar+memory – RoPE+memory	+50.3	+51.9	[+18.8, +92.5]	20/20

Table 1: Matched contrasts across the 120-cell, 1B-token factorial. Values are percentage-point changes in teacher-forced five-token accuracy at 64K. Each row holds all other factorial axes fixed. “Positive” excludes one numerical tie in the Polar–NoPE comparison.

Group	Model	Retrieval (%)			BABILong (%)		BPB		
		Tok. 2K	Tok. 256K	Exact 256K	2K	256K	2K	256K	ratio
Matched	NoPE	99.2	0.6	0.0	55	0	1.432	8.097	5.65×
Matched	Polar	98.9	34.4	9.0	52	28	1.451	1.825	1.26×
Matched	RoPE	88.2	0.0	0.0	64	14	1.439	2.512	1.75×
Raven/AdamW	Atma-Raven-Titans	43.1	10.0	0.0	56	38	1.525	1.551	1.02×
Raven/AdamW	Raven Native	41.1	20.3	0.0	58	46	1.581	1.597	1.01×

Table 2: Long-context endpoints. Retrieval averages tasks, suites, and depths; “Tok.” is teacher-forced target-token accuracy and “Exact” requires all five target tokens to be correct. BABILong is macro exact match. BPB averages three fixed-target datasets; lower is better. Bold is best within each comparison group.

5.2 A multi-objective frontier

Table 2 reports extrapolation endpoints. At 2K, NoPE and Polar achieve nearly perfect target-token accuracy. At 256K, NoPE and RoPE fall to 0.6% and 0.0%, while Polar retains 34.4% averaged across tasks, suites, and depths. Exact five-token accuracy gives a stricter boundary: Polar reaches 9.0% overall at 256K, entirely from synthetic haystacks (18.0% synthetic; 0.0% FinePDFs). At 64K, Polar obtains 65.7% exact accuracy on synthetic contexts but 0.0% on FinePDFs, despite 16.3% target-token accuracy there. We therefore claim retained target signal and exact synthetic retrieval, not successful real-text retrieval at those lengths. Full token, exact, depth, and haystack matrices are in Appendix F.

Table 3 separates target-token accuracy by haystack and length; Appendix F reports the corresponding exact-sequence values. Polar also limits mean BPB degradation to 1.26× at 256K, compared with 5.65× for NoPE and 1.75× for RoPE. This stability accompanies a short-context cost: mean accuracy over eight 2K controls is 43.29% for Polar, 45.15% for NoPE, and 44.60% for RoPE. Figure 3 shows that the gap is task-dependent rather than uniform. The 1.86-point mean gap to NoPE prevents a blanket quality claim. Exact per-task values are in Appendix G.

Table 4 makes the quality–systems trade-off explicit. Raven Native reaches 46% BABILong at 256K and keeps BPB nearly flat, while its fixed recurrent state avoids a sequence-length-dependent KV cache. At 128K, Atma-Raven-Titans decodes at 2.27 ms/token versus 16.3 ms/token for Polar; Polar instead has higher synthetic retrieval and training MFU. The BABILong samples are small: simple Wilson intervals at 256K are 20.1–37.5% for Polar (28/100) and 36.6–55.7% for Raven Native (46/100). Because Raven also changes model family and optimizer, these rows locate practical operating points rather than isolate one causal mechanism.

Model	Haystack	2K	4K	8K	16K	32K	64K	128K	256K
Polar	Synthetic	99.6	98.4	98.5	97.0	94.9	92.3	80.3	68.4
	FinePDFs	98.6	86.3	77.9	49.1	40.7	16.3	3.2	0.3
NoPE	Synthetic	99.5	98.7	98.9	94.9	70.3	26.5	10.2	1.2
	FinePDFs	99.3	52.9	0.0	0.0	0.0	0.0	0.0	0.0
RoPE	Synthetic	81.7	44.9	33.1	0.8	6.8	0.1	0.0	0.0
	FinePDFs	94.8	4.7	0.2	0.0	0.0	0.0	0.0	0.0
Raven Native	Synthetic	55.9	57.7	51.4	48.3	44.3	40.9	38.9	35.1
	FinePDFs	22.1	15.4	8.2	3.3	5.3	2.9	3.7	5.4

Table 3: Zero-shot retrieval accuracy (%) by haystack type. Values average teacher-forced token accuracy over needle depths. The recurrent hybrid is omitted for space and reported in Appendix F.

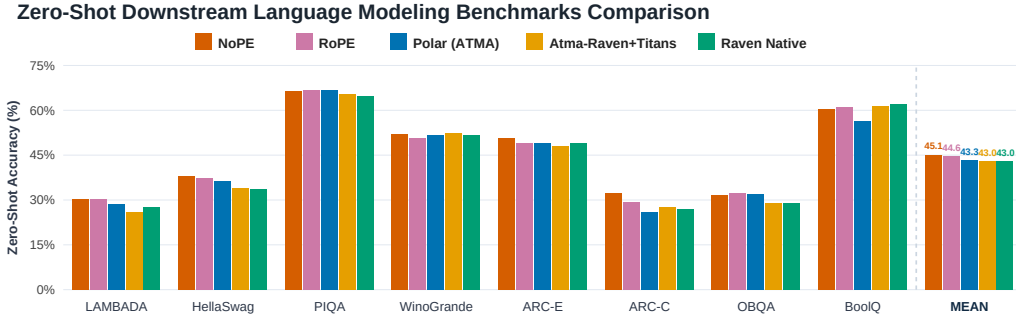


Figure 3: Zero-shot accuracy on eight 2K language-modeling controls (22,939 examples per model). The rightmost bars are unweighted task means. Exact values and metric definitions are in Table 11.

Group	Model	Base mean (%)	128K prefill (s)	Decode (ms/token)	Peak alloc. (GiB)	Train MFU (% 2K)
Matched	NoPE	45.1	1.47	16.4	39.40	42.77
Matched	Polar	43.3	1.62	16.3	39.41	40.44
Matched	RoPE	44.6	1.47	16.5	39.41	42.45
Raven/AdamW	Atma-Raven-Titans	43.1	1.00	2.27	8.62	30.92
Raven/AdamW	Raven Native	43.0	1.36	3.27	6.46	21.53

Table 4: Short-context and systems trade-offs. Serving uses one sequence and one timing sample on one L40S, so the values are descriptive rather than uncertainty estimates.

6 Checkpoint Variability Audit

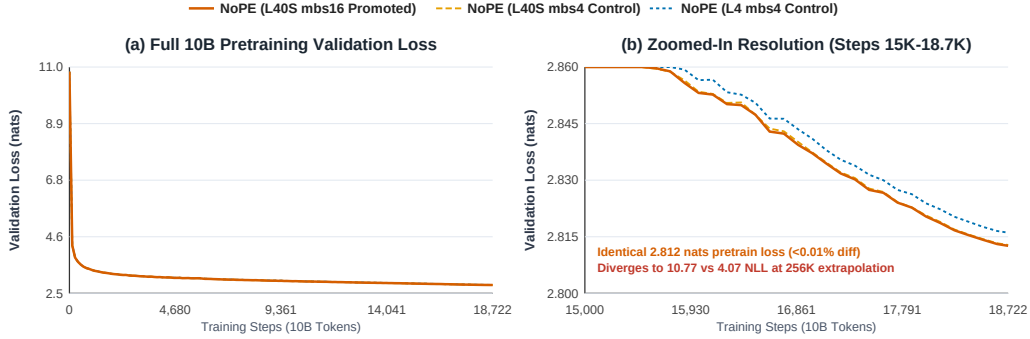


Figure 4: NoPE 2K validation trajectories are nearly coincident, yet the checkpoints’ 256K NLL values are 10.77, 9.90, and 4.07. The two L40S runs differ in microbatch size; the L4 run differs in device class.

Figure 4 presents a post-hoc audit: nearly identical short-context convergence can hide substantially different long-context behavior. Three NoPE runs finish at 2.8126, 2.8128, and 2.8160 validation nats, but their 256K NLLs are respectively 10.77 (L40S, microbatch 16), 9.90 (L40S, microbatch 4), and 4.07 (L4, microbatch 4). The runs were not initialized from a recorded common seed, so this comparison measures checkpoint-to-checkpoint variability, not a device-class treatment effect.

6.1 Diagnostic journey

The finding emerged unintentionally during an infrastructure transition rather than a planned device comparison. Stage I and the first 10B NoPE control ran on an L4 with microbatch 4 and produced 4.07 nats at 256K. Moving the promoted runs to L40S with microbatch 16 preserved the 2K trajectory but yielded 10.77 nats at 256K. Re-evaluation under PyTorch 2.12 and 2.13 left both checkpoints unchanged; an L40S microbatch-4 re-train still reached 9.90, and deterministic 2K fixtures showed no gradient or loss discrepancy. These checks exclude evaluation-version mismatch and microbatch size alone in the tested runs. They cannot distinguish random initialization, data-order effects, hardware-dependent kernels, or their interactions. All five headline models were trained in the same L40S environment, so the main model comparison remains environment-matched. Appendix J records the audit sequence.

FEA Structural Stress Probing: Local Block Secant Gains G Across Length (2K -> 256K)

Polar (ATMA): Flat Gains Across Length									RoPE: Representational Attenuation									NoPE (L40S mbs16): Block 7 Yield Locus								
	2K	4K	8K	16K	32K	64K	128K	256K		2K	4K	8K	16K	32K	64K	128K	256K		2K	4K	8K	16K	32K	64K	128K	256K
B0	0.86	0.86	0.86	0.86	0.86	0.86	0.86	0.86	B0	0.94	0.94	0.94	0.94	0.94	0.94	0.94	0.94	B0	0.91	0.91	0.91	0.91	0.91	0.91	0.91	0.91
B1	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	B1	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	B1	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88
B2	0.96	0.95	0.95	0.95	0.96	0.96	0.96	0.96	B2	1.01	1.00	0.99	0.98	0.98	0.98	0.98	0.98	B2	1.02	1.02	1.02	1.02	1.02	1.03	1.03	1.03
B3	0.90	0.89	0.89	0.89	0.88	0.88	0.88	0.87	B3	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	B3	0.98	0.98	0.98	0.97	0.97	0.97	0.97	0.97
B4	0.87	0.87	0.86	0.86	0.86	0.85	0.85	0.85	B4	1.05	1.07	1.08	1.07	1.06	1.04	1.03	1.03	B4	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.95
B5	0.86	0.85	0.85	0.84	0.84	0.84	0.84	0.84	B5	0.94	0.93	0.93	0.93	0.93	0.93	0.93	0.92	B5	0.94	0.94	0.94	0.94	0.95	0.96	0.96	0.96
B6	1.02	1.02	1.03	1.05	1.07	1.09	1.10	1.11	B6	1.09	1.06	1.04	1.02	1.01	1.01	1.00	1.00	B6	1.13	1.11	1.09	1.09	1.08	1.09	1.09	1.10
B7	0.98	0.97	0.97	0.97	0.97	0.97	0.97	0.97	B7	1.01	1.01	1.00	1.00	1.00	1.00	0.99	0.99	B7	1.02	1.02	1.02	1.03	1.09	1.20	1.31	1.38
B8	0.96	0.96	0.95	0.95	0.95	0.95	0.95	0.94	B8	1.01	1.00	1.00	1.00	0.99	0.99	0.99	0.99	B8	1.01	1.01	1.01	1.00	1.00	1.00	1.00	0.99
B9	0.93	0.92	0.92	0.92	0.92	0.92	0.92	0.91	B9	0.98	0.98	0.98	0.97	0.97	0.97	0.97	0.96	B9	0.99	0.99	0.99	0.98	0.98	0.98	0.98	0.99
B10	1.10	1.10	1.10	1.10	1.10	1.10	1.10	1.09	B10	1.09	1.08	1.07	1.06	1.06	1.05	1.04	1.04	B10	1.16	1.14	1.12	1.12	1.12	1.11	1.11	1.11
B11	1.02	1.02	1.02	1.02	1.02	1.01	1.01	1.01	B11	1.02	1.02	1.02	1.01	1.01	1.01	1.00	1.00	B11	1.04	1.03	1.03	1.02	1.01	1.01	1.00	1.00
B12	1.02	1.01	1.01	1.01	1.01	1.00	1.00	1.00	B12	1.02	1.02	1.01	1.01	1.01	1.00	1.00	1.00	B12	1.04	1.04	1.03	1.02	1.01	1.01	1.01	1.00
B13	1.01	1.01	1.00	1.00	1.00	1.00	1.00	1.00	B13	1.02	1.01	1.01	1.00	1.00	1.00	1.00	1.00	B13	1.04	1.04	1.06	1.08	1.09	1.07	1.06	1.06
B14	1.43	1.44	1.45	1.44	1.41	1.39	1.38	1.37	B14	1.32	1.31	1.29	1.28	1.27	1.26	1.25	1.25	B14	1.39	1.37	1.35	1.35	1.30	1.26	1.24	1.23
B15	1.23	1.23	1.22	1.21	1.20	1.19	1.18	1.17	B15	1.21	1.20	1.19	1.19	1.18	1.17	1.17	1.16	B15	1.18	1.18	1.17	1.14	1.12	1.11	1.09	1.08

Figure 5: Sampled block secant gains across context length. The probe localizes the L40S NoPE checkpoint’s change near blocks 6–7, while Polar remains comparatively flat. A sampled gain is a directional diagnostic, not a tight global Lipschitz estimate.

Figure 5 localizes the checkpoint difference with a sampled block-level stress probe. The L40S NoPE checkpoint first amplifies the perturbations around blocks 6–7, whereas the L4 checkpoint redistributes them more weakly downstream. This shows that the difference is stored in the trained checkpoints rather than introduced by the evaluation runtime. It does not establish why the training trajectories diverged. The defensible conclusion is that short-context convergence did not predict long-context behavior in these runs. Extrapolation studies should therefore report seeds, kernels, device classes, and extended-context audits; causal hardware claims require replicated, seed-paired interventions. Additional diagnostics are in Appendix I.

6.2 What this audit cannot answer

The finding opens several concrete questions that the present compute budget and page limit do not resolve:

1. Does the checkpoint divergence replicate under seed-paired device interventions, across software stacks, and at larger model scales?
2. Which training-level numerical difference—kernel selection, reduction or accumulation order, or optimizer-state evolution—is amplified into the divergent checkpoint, and can an early diagnostic predict it?
3. Does Polar’s 0.3% natural-text retrieval at 256K improve with longer or mixed-domain training, or reflect a synthetic-to-natural induction bias?
4. How much of Raven’s 256K BABILong lead comes from recurrent state, sparse routing, or AdamW, given that the Muon schedule did not converge?

These questions delimit the current evidence rather than extend its claims. Resolving them requires interventions that separately vary architecture, optimizer, data, and execution environment.

7 Limitations and Conclusion

This study uses one model scale, one 2K pretraining regime, and no seed replication sufficient for population-level architectural uncertainty. The Stage I factorial demonstrates robustness across nuisance-factor settings, not across random seeds. Teacher-forced retrieval, custom BABILong adaptation, and single-sample serving measurements answer different questions

and should not be collapsed into one score. Polar has zero exact FinePDFs retrieval at 64K and 256K despite retaining partial target-token signal, its theoretical bounds require learned-margin assumptions not yet certified for every head, and Raven is not an optimizer-matched ablation. The documented pre-integration Polar design tests are not a substitute for network-level component ablations, and the present comparisons do not cover every contemporary length-scaling method.

Within those limits, ATMA establishes a reproducible operating point for bounded full-context attention plus recurrent memory. Polar materially extends target-token signal, preserves exact synthetic retrieval, and stabilizes long-document likelihood while remaining competitive on short-context tasks. Raven illustrates the BABILong and serving advantages of compressed recurrent state. The checkpoint audit further shows why length-extrapolation claims require evaluation far beyond the training window and careful reporting of run meta-data. We release configurations, logs, checkpoint manifests, benchmark matrices, and numerical proof artifacts so that each reported comparison can be audited.

AI Use Statement

Generative AI tools were used in this revision to edit and shorten the manuscript, improve document structure, and reaggregate existing archived results. They were not used to generate experimental datasets, train models, or run new evaluations in this revision. The authors manually compare all AI-assisted mathematical text with the original derivations and proof artifacts, verify empirical values against source logs, and take responsibility for every claim and artifact in the submission. This statement describes the present version as required by the ICLR 2027 policy.

Reproducibility Statement

Model definitions and training settings are described in Sections 3–4. Appendix D specifies evaluation protocols and denominators, Appendices E–I provide detailed tables and diagnostics. The anonymous artifact at <https://anonymous.4open.science/r/atma> contains configurations, logs, checkpoint manifests, benchmark payloads, and machine-checked proof skeletons.

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A Theoretical Framing

These are sufficient-condition results. They establish noise-odds and norm bounds under explicit assumptions. They do not establish distributional invariance or replace the empirical evidence.

A.1 Extreme-value null floor

Assumption 1 (Sub-Gaussian noise scores). For irrelevant keys $j \in \mathcal{N}_n$, $|\mathcal{N}_n| \leq n$, scores X_j are mean-zero and v^2 -sub-Gaussian: $\mathbb{E} \exp(tX_j) \leq \exp(v^2 t^2/2)$ for all $t \in \mathbb{R}$.

Let the Polar null floor be $\nu_n = b + s\sqrt{\log(n+1)}$ with $b, s \geq 0$.

Proposition 1 (Noise exceedances). Under Assumption 1,

$$\mathbb{E} \left[\sum_{j \in \mathcal{N}_n} \mathbf{1}\{X_j > \nu_n\} \right] \leq n \exp\left(-\frac{\nu_n^2}{2v^2}\right). \quad (5)$$

If $s \geq \sqrt{2}v$ this is $O(1)$; if the inequality is strict it decays polynomially.

Proof. Apply the sub-Gaussian tail bound to each key and sum. Since $b \geq 0$, $\nu_n^2 \geq s^2 \log(n+1)$. $\square$

Unlike TDA, Polar retains sub-threshold keys. With temperature $\theta_n = 1 + a \log n$, define the unnormalized noise-to-null odds $R_n = \sum_{j \in \mathcal{N}_n} \exp(\theta_n(X_j - \nu_n))$.

Proposition 2 (Vanishing null odds). Suppose $s > c > \sqrt{2}v$ and the null floor is $s\sqrt{\log(n+1)}$ up to a nonnegative offset. With probability tending to one,

$$R_n \leq n \exp\left(-(1+a \log n)(s-c)\sqrt{\log n}\right). \quad (6)$$

For $a > 0$, the bound converges to zero faster than any inverse polynomial.

Proof. A union bound gives $\Pr[\max_j X_j > c\sqrt{\log(n+1)}] \leq n(n+1)^{-c^2/(2v^2)} \rightarrow 0$. On the complementary event, every noise score is at least $(s-c)\sqrt{\log n}$ below the floor. Bound each summand and sum over at most n keys. $\square$

A.2 Bounded channels and recurrent state

Proposition 3 (Bounded Polar readout). For $c_i = s_i / \max(\|s_i\|_2, \varepsilon)$, $\|c_i\|_2 \leq 1$. For $\mu_i = \tanh(\beta \log(1 + m_{\text{eff},i}))$, $\mu_i \in [0, 1)$ when $\beta, m_{\text{eff},i} \geq 0$. A distribution uniform over m real keys has participation ratio m .

Proof. The first two claims follow from the denominator and range of $\tanh$. Uniform weights satisfy $(\sum_{j=1}^m (1/m)^2)^{-1} = m$. $\square$

Assumption 2 (Contractive memory). For every t , $\|q_t\|_2 = \|k_t\|_2 = 1$, $\|v_t\|_2 \leq V$, $0 \leq \beta_t \leq 1$, and $0 \leq \gamma_t \leq \bar{\gamma} < 1$.

Proposition 4 (Length-bounded state). For recurrence equation 4,

$$\|M_t\|_F \leq \bar{\gamma}^t \|M_0\|_F + \frac{V}{1 - \bar{\gamma}}. \quad (7)$$

Proof. $I - \beta_t k_t k_t^\top$ has spectral norm at most one and $\|\beta_t v_t k_t^\top\|_F \leq V$. Thus $\|M_t\|_F \leq \bar{\gamma}\|M_{t-1}\|_F + V$; unrolling gives the result. $\square$

The Lean artifact checks the finite-sum, exponential-monotonicity, bounded-channel composition, and invariant-set induction steps. It takes the sub-Gaussian tail, high-probability margin, and one-step contraction as premises. It is therefore a machine-checked proof skeleton, not end-to-end formal verification. The learned conditions $b \geq 0$ and $s > c > \sqrt{2}v$ require an empirical head-wise audit before they can be asserted for a trained checkpoint.

B Regularization Modes

The factorial grid varies a representation regularizer applied to the model’s signature stream. For every non-baseline mode, the sweep uses the same weight $\alpha_{\text{sig}} = 0.01$ and optimizes

$$\mathcal{L} = (1 - \alpha_{\text{sig}})\mathcal{L}_{\text{LM}} + \alpha_{\text{sig}}\mathcal{L}_{\text{reg}} + \lambda_{\text{dist}}\mathcal{L}_{\text{dist}}. \quad (8)$$

The weak and strong modes follow the two SIGReg families introduced by Weak-SIGReg (Akbar, 2026) and LeJEPA’s strong SIGReg objective (Balestriero & LeCun, 2025). The discrete and zipfian modes are local variants that combine those ideas with stronger geometric priors. Table 5 defines all five sweep settings.

Mode	Meaning in the sweep
baseline	No signature-stream regularization; equivalently, $\alpha_{\text{sig}} = 0$.
weak	Weak SIGReg: center a random sketch of the token representations and match its covariance to the identity, $\ \text{Cov}(Sx) - I\ _F$. This encourages decorrelated, unit-variance features at low cost.
strong	Strong SIGReg: project representations onto random one-dimensional directions and match their empirical characteristic functions to the standard Gaussian characteristic function over a fixed frequency grid.
discrete	Local variant: normalize each representation to the sphere of radius $\sqrt{d}$, then apply weak covariance matching. This preserves whitening pressure while encouraging fixed-norm, discrete-code-like directions.
zipfian	Local variant: normalize directions for an orthogonality penalty, then match sorted representation magnitudes to a Zipf profile r^{-1} . This biases the signature stream toward heavy-tailed feature use rather than uniform energy allocation.

Table 5: Regularization modes in the 120-cell factorial sweep. Weak and strong are SIGReg-style covariance and distribution-matching objectives; discrete and zipfian are exploratory local variants.

C Polar Attention Implementation

C.1 Design provenance and scope

Before integration, the Polar choices were developed by falsifying simpler alternatives in a standalone synthetic sandbox. Additive soft counts accumulated small per-key leakage with sequence length; thresholds based on a fixed number of standard deviations retained a constant fraction of noise keys; a fixed null logit was eventually overtaken by the maximum noise score; and an unbounded $\log m$ magnitude exposed the output projection to values beyond its training range. These failures motivated participation ratio, the $\sqrt{\log n}$ null floor, and the bounded $\tanh(\beta \log(1 + m))$ map. The standalone prototype was subsequently removed and is not part of the released artifact. We therefore use these tests as design provenance, not as model-scale component ablations or independently reproducible benchmark evidence. The integrated equations, materialized oracle, streaming implementation, and parity tests described below are retained.

C.2 Numerical oracle

The materialized PyTorch path forms the masked real-key logits and virtual null logit in FP32, applies a single softmax, and computes the direction and magnitude channels from the resulting weights. It is used as a numerical oracle for forward and backward tests, not as the production long-context path, because materializing the $T \times T$ score matrix is memory-prohibitive at the evaluated lengths.

C.3 Streaming sufficient statistics

The Triton kernel follows the online-max strategy used by FlashAttention-style kernels. For one query row, it streams K, V blocks and retains only four FP32 statistics: the running maximum M , real-key exponential sum L , squared exponential sum Q_2 , and unnormalized value sum S . If a new block changes the row maximum, $\alpha = \exp(M_{\text{old}} - M_{\text{new}})$ rescales each statistic as shown in Table 6.

State	Online update	Use after streaming
M	$M \leftarrow \max(M, \max a)$	Stable common logit origin
L	$L \leftarrow \alpha L + \sum_j p_j$	Real-key softmax mass
Q_2	$Q_2 \leftarrow \alpha^2 Q_2 + \sum_j p_j^2$	$n_{\text{eff}} = L^2 / Q_2$
S	$S \leftarrow \alpha S + \sum_j p_j v_j$	Direction numerator

Table 6: Streaming state for one Polar query row, where a is the current logit block and $p_j = \exp(a_j - M_{\text{new}})$. The squared accumulator must rescale by α^2 .

After the real keys are consumed, the virtual null is folded in as one additional softmax entry. Let $p_0 = \exp(t_i v_i - M)$ and $Z = L + p_0$. The kernel obtains

$$c_i = \text{normalize}(S + p_0 v_{\text{null}}), \quad n_{\text{eff},i} = \frac{L^2}{Q_2}, \quad m_{\text{eff},i} = n_{\text{eff},i} \frac{L}{Z}. \quad (9)$$

The common factor $1/Z$ cancels under radial normalization of the direction numerator. The production kernel evaluates the magnitude with $2\sigma(2x) - 1 = \tanh x$ and saves (M, L, Q_2, S) for the backward pass. Listing 1 gives the algorithmic core; pointer arithmetic, launch metadata, and backward code are omitted.

Listing 1: Schematic core of the block-streamed Triton forward pass.

```

@triton.jit
def polar_fwd_core(q, K, V, n_i, temp, nullv, beta,
    BLOCK_N: tl.constexpr, DK: tl.constexpr):
    M = tl.full([BLOCK_M], -1e38, tl.float32)
    L = tl.zeros([BLOCK_M], tl.float32)
    Q2 = tl.zeros([BLOCK_M], tl.float32)
    S = tl.zeros([BLOCK_M, DK], tl.float32)

    for start in range(0, n_i, BLOCK_N):
        k, v, valid = load_kv_block(K, V, start, n_i)
        a = tl.where(valid, temp[:, None] * dot(q, k), -1e38)
        M_new = tl.maximum(M, tl.max(a, axis=1))
        alpha = tl.exp(M - M_new)
        p = tl.where(valid, tl.exp(a - M_new[:, None]), 0.0)
        L = alpha * L + tl.sum(p, axis=1)
        Q2 = alpha * alpha * Q2 + tl.sum(p * p, axis=1)
        S = alpha[:, None] * S + tl.dot(p, v)
        M = M_new

    # Fold in the learned virtual null as one extra entry.
    M_new = tl.maximum(M, temp * nullv)
    alpha = tl.exp(M - M_new)
    L, Q2, S = alpha * L, alpha * alpha * Q2, alpha[:, None] * S
    p_null = tl.exp(temp * nullv - M_new)
    Z = L + p_null

    direction = normalize(S + p_null[:, None] * v_null)
    n_eff = L * L / tl.maximum(Q2, eps)
    m_eff = n_eff * L / tl.maximum(Z, eps)
    magnitude = 2 * tl.sigmoid(2 * beta * tl.log(1 + m_eff)) - 1

```

```
save_for_backward(M_new, L, Q2, S)
```

This removes the quadratic score-matrix storage. Full-context prefill still performs quadratic query-key arithmetic. The kernel is cross-checked against the materialized PyTorch oracle in FP32 accumulation before lower-precision outputs are accepted.

D Evaluation Protocols

Stage I contains 120 approximately 370–378M-parameter runs trained for 1,900 steps of 524,288 tokens at sequence length 2,048 on FineWeb-Edu. Stage II contains three matched attention models trained for 18,722 such steps (9.816B tokens). The five promoted checkpoints and ten open-weight controls are evaluated with checkpoint-exact forward paths. Stage I cells vary factorial settings and are not independent seed replicates.

D.1 Raven optimizer control

The Raven rows use AdamW optimizer because the attempted matched-schedule control did not produce a usable baseline. In the 1B-token Raven Native pilot with the Atma Muon schedule, validation loss decreased from 10.83 to 3.76 nats by step 625, then rose to 6.88 at step 1,125 and ended at 4.62; teacher-forced retrieval was 0% at every tested length from 2K through 64K. The reported Raven Native and Atma-Raven-Titans checkpoints therefore use AdamW and remain a separate model-family/optimizer group. This is evidence about the tested Muon schedule, not a general claim that Raven cannot converge under any Muon tuning.

Retrieval crosses passkey and NIAH tasks, synthetic and FinePDFs haystacks, eight lengths from 2K through 256K, three depths, and 50 paired trials per cell. The scorer appends the complete five-token target and evaluates each position conditioned on the preceding ground-truth target tokens. Token accuracy is the fraction of correct per-position argmax predictions; exact accuracy requires all five positions to be correct. Both are teacher-forced diagnostics, not autoregressive generation. An older free-generation protocol was retired because all base checkpoints scored uniformly zero. BABILong uses answer-only fine-tuning on tasks qa1–qa10 at 0K/1K/2K and greedy exact match on ten held-out rows per task and length. Base-task controls use LAMBADA, HellaSwag, PIQA, WinoGrande, ARC-Easy, ARC-Challenge, OpenBookQA, and BoolQ. Fixed-target BPB uses FinePDFs, PG-19, and Proof-Pile. All reported processes completed without failed, OOM, unsupported, missing, or null cells.

For proportions we report simple Wilson intervals where they aid interpretation. These intervals treat trials as binomial observations and do not exploit the paired design or account for correlations induced by shared templates. The archived retrieval payloads retain aggregate cell counts but the BABILong payloads do not retain per-model correctness vectors, so paired bootstrap or McNemar tests cannot be reconstructed without rerunning or recovering the original predictions.

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E Complete Factorial Sweep

Table 7 retains all 120 factorial cells; the main paper reports only the recipe transitions needed to justify promotion.

Table 7: Complete Results of the 120-Run Ablation Sweep. All models are 370M parameters, trained on FineWeb-Edu for 1B tokens ($seq_len = 2048$). Evaluated from 2K to 64K context length. Clean NLL is measured on coherent FinePDFs documents; junk NLL is measured on the concatenated validation stream. NLL is in nats/token (lower is better), Needle is greedy per-digit accuracy in % (higher is better), and MFU is training model flops utilization in %.

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
POLAR	baseline	Off	Off	Off	3.323	2.83	3.60	3.27	5.68	96.3	0.0	40.1
POLAR	baseline	Off	Off	On	3.343	2.83	2.66	3.30	4.74	67.5	0.0	39.6
POLAR	baseline	Off	On	Off	3.169	2.70	1.96	3.11	3.13	91.3	92.5	35.6
POLAR	baseline	Off	On	On	3.174	2.71	2.01	3.12	3.13	51.3	30.0	35.6
POLAR	baseline	On	Off	Off	3.332	2.81	5.56	3.28	6.54	85.0	0.0	33.3
POLAR	baseline	On	Off	On	3.361	2.85	2.89	3.35	4.97	91.3	2.5	34.5
POLAR	baseline	On	On	Off	3.178	2.72	1.98	3.12	3.14	73.8	58.8	31.8
POLAR	baseline	On	On	On	3.182	2.73	1.97	3.12	3.14	56.3	21.3	31.2
POLAR	weak	Off	Off	Off	3.333	2.82	2.86	3.28	5.35	88.8	0.0	39.0
POLAR	weak	Off	Off	On	3.348	2.90	3.76	3.38	5.24	93.8	0.0	37.9
POLAR	weak	Off	On	Off	3.179	2.74	1.95	3.12	3.14	70.0	51.3	36.2
POLAR	weak	Off	On	On	3.187	2.74	1.99	3.13	3.15	41.3	41.3	36.2
POLAR	weak	On	Off	Off	3.335	2.82	3.47	3.28	5.56	91.3	0.0	32.8
POLAR	weak	On	Off	On	3.353	2.82	2.58	3.32	4.81	96.3	1.3	32.8
POLAR	weak	On	On	Off	3.178	2.74	1.95	3.12	3.14	33.8	32.5	30.7
POLAR	weak	On	On	On	3.183	2.74	1.94	3.13	3.15	81.3	78.8	30.7
POLAR	strong	Off	Off	Off	3.338	2.85	3.79	3.28	5.83	98.8	1.3	37.3
POLAR	strong	Off	Off	On	3.360	2.85	2.99	3.32	5.01	91.3	0.0	37.9
POLAR	strong	Off	On	Off	3.172	2.72	1.96	3.11	3.13	80.0	52.5	35.8

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Table 7: Complete Results of the 120-Run Ablation Sweep (continued).

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
POLAR	strong	Off	On	On	3.181	2.72	1.94	3.12	3.15	76.3	68.8	34.6
POLAR	strong	On	Off	Off	3.337	2.87	3.22	3.28	5.52	83.8	1.3	32.8
POLAR	strong	On	Off	On	3.347	2.82	2.64	3.30	4.88	97.5	16.3	32.2
POLAR	strong	On	On	Off	3.179	2.73	1.94	3.12	3.14	86.3	83.8	30.9
POLAR	strong	On	On	On	3.182	2.72	1.94	3.12	3.14	25.0	23.8	30.9
POLAR	discrete	Off	Off	Off	3.338	2.82	2.80	3.29	5.12	93.8	2.5	37.9
POLAR	discrete	Off	Off	On	3.349	2.87	2.90	3.33	5.25	93.8	6.3	37.9
POLAR	discrete	Off	On	Off	3.184	2.75	2.03	3.12	3.14	76.3	73.8	36.4
POLAR	discrete	Off	On	On	3.178	2.71	1.92	3.12	3.15	31.3	18.8	35.8
POLAR	discrete	On	Off	Off	3.333	2.80	4.13	3.28	6.12	97.5	0.0	32.6
POLAR	discrete	On	Off	On	3.351	2.87	2.72	3.33	4.89	85.0	2.5	33.3
POLAR	discrete	On	On	Off	3.189	2.74	1.96	3.13	3.15	87.5	53.8	31.8
POLAR	discrete	On	On	On	3.192	2.74	2.07	3.13	3.15	21.3	26.3	31.7
POLAR	zipfian	Off	Off	Off	3.338	2.85	4.83	3.28	6.55	96.3	11.3	37.9
POLAR	zipfian	Off	Off	On	3.350	2.83	2.58	3.32	4.63	88.8	3.8	39.4
POLAR	zipfian	Off	On	Off	3.171	2.70	1.92	3.11	3.13	52.5	63.8	36.2
POLAR	zipfian	Off	On	On	3.180	2.75	1.95	3.12	3.15	46.3	40.0	35.2
POLAR	zipfian	On	Off	Off	3.340	2.83	3.77	3.28	5.64	98.8	1.3	33.8
POLAR	zipfian	On	Off	On	3.352	2.85	3.13	3.36	5.23	90.0	0.0	33.5
POLAR	zipfian	On	On	Off	3.177	2.72	1.94	3.12	3.14	91.3	67.5	31.8
POLAR	zipfian	On	On	On	3.192	2.74	2.00	3.13	3.15	40.0	26.3	31.8
NOPE	baseline	Off	Off	Off	3.224	2.76	3.71	3.16	6.68	97.5	1.3	41.5
NOPE	baseline	Off	Off	On	3.219	2.75	2.76	3.17	5.49	92.5	10.0	36.5
NOPE	baseline	Off	On	Off	3.140	2.66	2.34	3.09	3.23	97.5	16.3	36.8
NOPE	baseline	Off	On	On	3.150	2.70	2.14	3.09	3.24	81.3	0.0	34.2
NOPE	baseline	On	Off	Off	3.209	2.74	2.99	3.15	6.13	90.0	18.8	32.5
NOPE	baseline	On	Off	On	3.212	2.77	2.90	3.17	5.65	81.3	0.0	30.3
NOPE	baseline	On	On	Off	3.149	2.67	2.35	3.10	3.28	80.0	16.3	30.0
NOPE	baseline	On	On	On	3.147	2.68	2.09	3.09	3.23	88.8	21.3	28.2
NOPE	weak	Off	Off	Off	3.214	2.77	3.10	3.16	6.34	95.0	0.0	40.9

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Table 7: Complete Results of the 120-Run Ablation Sweep (continued).

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
NOPE	weak	Off	Off	On	3.222	2.75	3.05	3.19	5.81	81.3	0.0	36.1
NOPE	weak	Off	On	Off	3.147	2.68	2.07	3.09	3.19	88.8	21.3	37.5
NOPE	weak	Off	On	On	3.142	2.68	2.12	3.08	3.23	82.5	0.0	33.7
NOPE	weak	On	Off	Off	3.215	2.75	4.87	3.16	12.48	95.0	5.0	32.0
NOPE	weak	On	Off	On	3.221	2.73	2.63	3.18	4.84	81.3	0.0	29.3
NOPE	weak	On	On	Off	3.147	2.67	2.48	3.09	3.23	97.5	1.3	30.8
NOPE	weak	On	On	On	3.148	2.68	2.14	3.09	3.24	93.8	1.3	28.4
NOPE	strong	Off	Off	Off	3.231	2.74	4.88	3.18	11.85	90.0	2.5	38.4
NOPE	strong	Off	Off	On	3.214	2.72	2.68	3.17	5.28	82.5	0.0	36.4
NOPE	strong	Off	On	Off	3.144	2.68	2.37	3.09	3.24	96.3	6.3	35.7
NOPE	strong	Off	On	On	3.150	2.70	2.10	3.10	3.25	93.8	7.5	33.8
NOPE	strong	On	Off	Off	3.207	2.72	3.13	3.15	6.20	73.8	0.0	31.4
NOPE	strong	On	Off	On	3.214	2.74	2.49	3.17	4.36	86.3	8.8	28.9
NOPE	strong	On	On	Off	3.146	2.67	2.24	3.09	3.23	83.8	17.5	29.1
NOPE	strong	On	On	On	3.147	2.68	2.16	3.09	3.28	96.3	0.0	27.5
NOPE	discrete	Off	Off	Off	3.202	2.75	3.33	3.15	6.23	85.0	1.3	38.8
NOPE	discrete	Off	Off	On	3.234	2.79	3.11	3.19	5.47	70.0	18.8	37.0
NOPE	discrete	Off	On	Off	3.145	2.68	2.14	3.09	3.27	98.8	16.3	36.0
NOPE	discrete	Off	On	On	3.145	2.68	2.10	3.09	3.25	95.0	3.8	34.3
NOPE	discrete	On	Off	Off	3.212	2.88	3.04	3.16	6.88	42.5	0.0	31.2
NOPE	discrete	On	Off	On	3.211	2.78	2.73	3.16	5.28	82.5	1.3	29.1
NOPE	discrete	On	On	Off	3.145	2.68	2.27	3.08	3.26	91.3	11.3	30.3
NOPE	discrete	On	On	On	3.149	2.68	2.17	3.08	3.32	83.8	0.0	28.6
NOPE	zipfian	Off	Off	Off	3.209	2.71	3.18	3.15	6.40	90.0	0.0	40.8
NOPE	zipfian	Off	Off	On	3.219	2.75	2.71	3.17	5.44	77.5	1.3	36.1
NOPE	zipfian	Off	On	Off	3.141	2.67	2.55	3.08	3.38	96.3	0.0	38.0
NOPE	zipfian	Off	On	On	3.150	2.69	2.11	3.09	3.25	90.0	3.8	33.7
NOPE	zipfian	On	Off	Off	3.207	2.77	3.32	3.15	7.09	82.5	0.0	31.3
NOPE	zipfian	On	Off	On	3.227	2.79	2.91	3.18	5.94	93.8	11.3	30.2
NOPE	zipfian	On	On	Off	3.146	2.68	2.39	3.10	3.25	96.3	21.3	29.6

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Table 7: Complete Results of the 120-Run Ablation Sweep (continued).

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
NOPE	zipfian	On	On	On	3.148	2.68	2.17	3.09	3.28	90.0	6.3	28.7
ROPE	baseline	Off	Off	Off	3.159	2.85	3.36	3.09	5.21	42.5	0.0	40.4
ROPE	baseline	Off	Off	On	3.163	2.86	3.79	3.17	5.86	15.0	0.0	37.2
ROPE	baseline	Off	On	Off	3.145	2.81	2.86	3.09	3.57	73.8	0.0	37.1
ROPE	baseline	Off	On	On	3.151	2.80	2.95	3.12	3.72	13.8	0.0	34.4
ROPE	baseline	On	Off	Off	3.163	3.00	3.47	3.10	5.46	38.8	0.0	33.2
ROPE	baseline	On	Off	On	3.167	2.88	4.06	3.16	5.92	0.0	0.0	30.2
ROPE	baseline	On	On	Off	3.153	2.73	2.45	3.09	3.57	75.0	0.0	31.4
ROPE	baseline	On	On	On	3.154	2.83	2.86	3.12	3.72	32.5	0.0	29.4
ROPE	weak	Off	Off	Off	3.158	2.88	3.46	3.10	5.36	46.3	0.0	41.4
ROPE	weak	Off	Off	On	3.163	2.84	3.96	3.18	5.93	12.5	0.0	37.8
ROPE	weak	Off	On	Off	3.149	2.71	2.46	3.09	3.56	37.5	0.0	36.5
ROPE	weak	Off	On	On	3.156	2.85	2.77	3.12	3.60	23.8	0.0	34.9
ROPE	weak	On	Off	Off	3.155	2.83	3.49	3.10	5.27	38.8	0.0	32.3
ROPE	weak	On	Off	On	3.157	3.05	4.12	3.18	5.86	8.8	0.0	30.5
ROPE	weak	On	On	Off	3.149	2.77	2.76	3.09	3.56	57.5	0.0	31.1
ROPE	weak	On	On	On	3.149	2.77	3.13	3.12	3.80	17.5	0.0	28.0
ROPE	strong	Off	Off	Off	3.165	2.78	3.34	3.10	5.27	30.0	0.0	39.0
ROPE	strong	Off	Off	On	3.169	2.90	3.68	3.19	5.80	31.3	0.0	36.3
ROPE	strong	Off	On	Off	3.158	2.85	2.63	3.10	3.50	51.3	0.0	35.9
ROPE	strong	Off	On	On	3.162	2.86	3.08	3.14	3.76	10.0	0.0	33.2
ROPE	strong	On	Off	Off	3.161	2.85	3.58	3.10	5.28	47.5	0.0	31.1
ROPE	strong	On	Off	On	3.165	2.85	4.03	3.19	5.90	2.5	0.0	29.2
ROPE	strong	On	On	Off	3.155	2.80	2.70	3.10	3.62	61.3	0.0	29.3
ROPE	strong	On	On	On	3.162	2.85	3.08	3.13	3.74	2.5	0.0	28.4
ROPE	discrete	Off	Off	Off	3.165	2.91	3.43	3.11	5.42	22.5	0.0	39.7
ROPE	discrete	Off	Off	On	3.158	2.84	4.05	3.15	5.90	3.8	0.0	36.9
ROPE	discrete	Off	On	Off	3.159	2.83	2.67	3.11	3.56	47.5	0.0	36.3
ROPE	discrete	Off	On	On	3.161	2.79	2.71	3.13	3.74	41.3	0.0	33.7
ROPE	discrete	On	Off	Off	3.171	2.79	3.45	3.12	5.36	51.3	0.0	31.5

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Table 7: Complete Results of the 120-Run Ablation Sweep (continued).

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
ROPE	discrete	On	Off	On	3.164	2.80	3.74	3.16	5.86	21.3	0.0	30.0
ROPE	discrete	On	On	Off	3.162	2.74	2.47	3.11	3.53	50.0	0.0	30.9
ROPE	discrete	On	On	On	3.154	2.79	2.68	3.11	3.67	18.8	0.0	28.9
ROPE	zipfian	Off	Off	Off	3.163	2.82	3.25	3.10	5.22	62.5	0.0	39.8
ROPE	zipfian	Off	Off	On	3.161	2.94	3.94	3.18	5.81	16.3	0.0	37.1
ROPE	zipfian	Off	On	Off	3.149	2.78	2.70	3.09	3.63	46.3	0.0	36.7
ROPE	zipfian	Off	On	On	3.151	2.80	2.63	3.11	3.68	2.5	0.0	33.9
ROPE	zipfian	On	Off	Off	3.161	2.78	3.38	3.10	5.32	47.5	0.0	32.6
ROPE	zipfian	On	Off	On	3.160	3.01	4.11	3.17	5.81	17.5	0.0	30.2
ROPE	zipfian	On	On	Off	3.149	2.73	2.39	3.09	3.55	62.5	0.0	29.8
ROPE	zipfian	On	On	On	3.152	2.79	2.99	3.12	3.78	26.3	0.0	28.5

F Retrieval Breakdown

NIAH Retrieval Depth-Sweep Heatmaps Across Lengths (2K -> 256K) and Depths (10%, 50%, 90%)

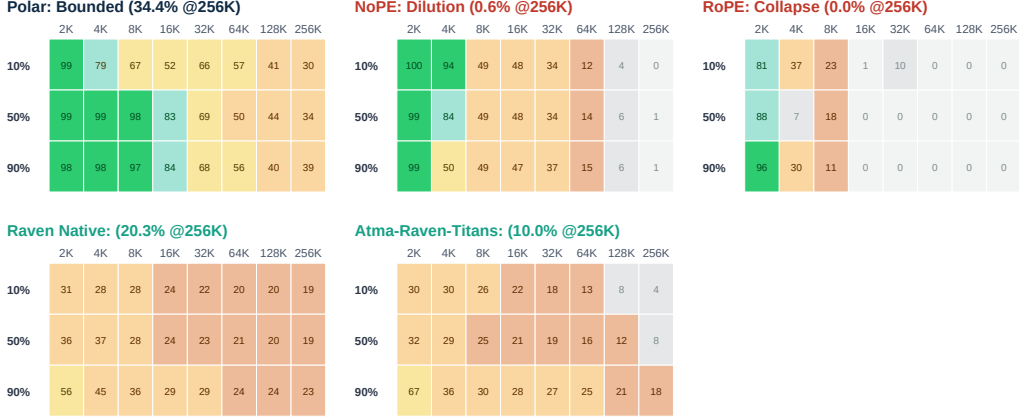


Figure 6: Retrieval depth sweeps for the five promoted models. Each cell is teacher-forced token accuracy.

Model	Haystack	2K	4K	8K	16K	32K	64K	128K	256K
Polar	Synthetic	99.6	98.4	98.5	97.0	94.9	92.3	80.3	68.4
	FinePDFs	98.6	86.3	77.9	49.1	40.7	16.3	3.2	0.3
NoPE	Synthetic	99.5	98.7	98.9	94.9	70.3	26.5	10.2	1.2
	FinePDFs	99.3	52.9	0.0	0.0	0.0	0.0	0.0	0.0
RoPE	Synthetic	81.7	44.9	33.1	0.8	6.8	0.1	0.0	0.0
	FinePDFs	94.8	4.7	0.2	0.0	0.0	0.0	0.0	0.0
Atma-Raven-Titans	Synthetic	64.8	60.7	51.1	46.9	41.7	35.8	25.9	19.3
	FinePDFs	23.2	2.6	0.7	0.2	1.4	0.5	1.6	0.7
Raven Native	Synthetic	55.9	57.7	51.4	48.3	44.3	40.9	38.9	35.1
	FinePDFs	22.1	15.4	8.2	3.3	5.3	2.9	3.7	5.4

Table 8: Zero-shot retrieval accuracy (%) by haystack type and context length, averaged over needle depths.

Figure 6 exposes depth-wise variation. Tables 8 and 9 separate partial target signal from exact five-token success. For Polar, the 64K exact rate is 65.7% on synthetic contexts (95% Wilson interval 60.1–70.8%) and 0/300 on FinePDFs (upper Wilson endpoint 1.3%). At 256K the corresponding exact rates are 18.0% (14.1–22.7%) and 0/300. These results support exact synthetic extrapolation and teacher-forced real-text signal, but not exact real-text retrieval.

F.1 Open-weight context

Table 10 reports the ten archived open-weight controls. They differ in training data, tokenizer, architecture, parameter count, and native context length; several were trained at lengths beyond 2K. They are therefore practical context, not matched length-extrapolation baselines and not evidence of a Polar advantage over the corresponding model families.

G Short-Context Quality and Systems

Table 11 gives the exact per-task controls shown in main-text Figure 3. Figure 7 shows the serving-length trend, and Table 12 summarizes the quality–systems trade-off.

Model	Haystack	Token accuracy (%)			Exact five-token accuracy (%)		
		2K	64K	256K	2K	64K	256K
Polar	Synthetic	99.6	92.3	68.4	96.3	65.7	18.0
	FinePDFs	98.6	16.3	0.3	93.0	0.0	0.0
NoPE	Synthetic	99.5	26.5	1.2	95.3	4.7	0.0
	FinePDFs	99.3	0.0	0.0	96.7	0.0	0.0
RoPE	Synthetic	81.7	0.1	0.0	42.3	0.0	0.0
	FinePDFs	94.8	0.0	0.0	77.3	0.0	0.0
Atma-Raven-Titans	Synthetic	64.8	35.8	19.3	8.0	0.3	0.0
	FinePDFs	23.2	0.5	0.7	4.7	0.0	0.0
Raven Native	Synthetic	55.9	40.9	35.1	2.0	0.0	0.0
	FinePDFs	22.1	2.9	5.4	0.0	0.0	0.0

Table 9: Teacher-forced retrieval under the two archived metrics. Token accuracy can expose partial target signal, whereas exact accuracy requires the entire five-token value to be predicted correctly under teacher forcing. Each haystack-length entry averages 300 sequences (two tasks, three depths, 50 trials).

Open-weight model	NLL 2K	NLL 64K	Needle 2K	Needle 64K
SmolLM2-360M	1.99	6.04	100.0	8.1
LFM2.5-230M	3.01	6.69	38.1	19.4
LFM2.5-350M	3.18	6.63	45.6	46.3
Qwen2.5-0.5B	2.03	1.53	100.0	38.1
Qwen3-0.6B-Base	1.95	1.40	100.0	95.0
Qwen3.5-0.8B-Base	1.94	1.33	100.0	100.0
Gemma-3-270M	2.48	1.90	96.2	24.4
Granite-4.0-350M	2.52	3.02	40.6	8.8
Granite-4.0-H-350M	2.22	2.06	75.0	54.4
Falcon-H1-0.5B	1.99	2.54	92.5	19.4

Table 10: Open-weight controls evaluated through 64K. NLL is fixed-target FinePDFs nats/token; Needle is teacher-forced target-token accuracy over 16 trials. These checkpoints are not matched for data, tokenizer, native context, or training budget.

Group	Model	LAMBADA	HellaSwag	PIQA	WinoGrande	ARC-E	ARC-C	OBQA	BoolQ	Mean
Matched	NoPE	30.16	37.88	66.27	51.93	50.53	32.44	31.60	60.37	45.15
Matched	RoPE	30.25	37.19	66.76	50.67	49.12	29.43	32.40	60.95	44.60
Matched	Polar	28.47	36.19	66.87	51.62	49.12	26.09	31.80	56.21	43.29
Raven/AdamW	Atma-Raven-Titans	26.00	34.12	65.56	52.49	48.07	27.76	29.00	61.41	43.05
Raven/AdamW	Raven Native	27.48	33.56	64.69	51.70	49.12	27.09	28.80	61.93	43.05

Table 11: Task-level zero-shot accuracy (%) on the eight short-context controls (22,939 examples per model). LAMBADA, WinoGrande, and BoolQ use standard accuracy; HellaSwag, PIQA, ARC-Easy, ARC-Challenge, and OpenBookQA use length-normalized accuracy. Mean is the unweighted average of the eight displayed metrics; bold is best within each comparison group.

H Systems Implementation and Optimization Breakdown

The inference stack uses separate kernels for full-context prefill, paged autoregressive decode, and recurrent-state updates. Table 13 distinguishes what each optimization changes. In particular, streaming removes quadratic score storage but not quadratic prefill arithmetic, while the recurrent Raven path changes the state representation itself and therefore has different scaling.

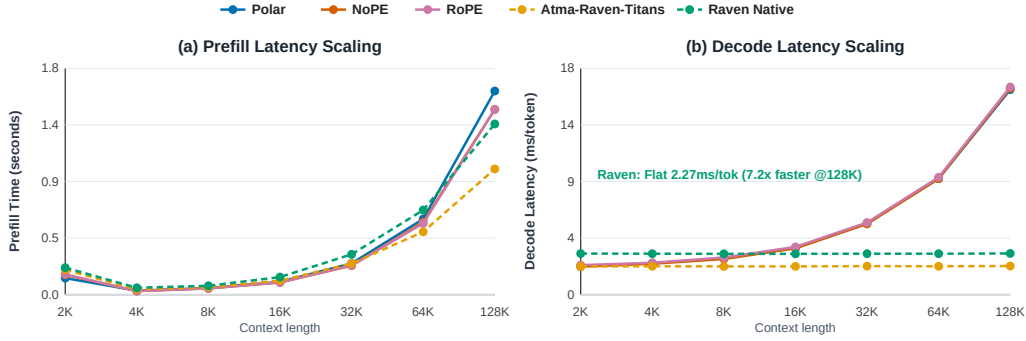


Figure 7: Single-sequence serving on one L40S. Attention decode scales with KV-cache length; recurrent Raven uses fixed state.

Group	Model	Base mean (%)	128K prefill (s)	Decode (ms/token)	Peak alloc. (GiB)	Train MFU (% , 2K)
Matched	NoPE	45.1	1.47	16.4	39.40	42.77
Matched	Polar	43.3	1.62	16.3	39.41	40.44
Matched	RoPE	44.6	1.47	16.5	39.41	42.45
Raven/AdamW	Atma-Raven-Titans	43.1	1.00	2.27	8.62	30.92
Raven/AdamW	Raven Native	43.0	1.36	3.27	6.46	21.53

Table 12: Short-context quality and serving trade-offs. Serving uses one sequence and one timing sample; it is descriptive rather than an uncertainty estimate.

Path	Optimization	Operational effect
Polar prefill	Block-streamed K, V tiles with online (M, L, Q_2, S) accumulation	Avoids materializing $T \times T$ scores; retains full-context $O(T^2)$ arithmetic.
Polar decode	Paged KV-cache kernel grouping the four query heads that share each KV head	Reuses KV loads across the GQA group; decode remains $O(T)$ in cache length.
Gated-delta step	Fused in-place decay, prediction, delta write, and readout	Avoids materializing per-step intermediate states and preserves $O(1)$ recurrent state size.
Ragged prefill	Packed prompts with grouped compatible tile shapes	Reduces padding work on heterogeneous prompt batches.
Compiler boundary	Recurrent scan and custom kernels exposed as opaque operations	Prevents compiler graph breaks around the stateful update while retaining explicit numerical tests.

Table 13: Implementation-level optimization breakdown. Complexity statements describe the sequence-mixing path and exclude shared MLP and embedding costs.

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H.1 Single-sequence scaling

Table 14 gives the underlying single-sequence measurements. On one NVIDIA L40S, attention decode increases from approximately 2.3 ms/token at 2K to 16.3–16.5 ms/token at 128K because every step reads a longer KV cache. The recurrent models remain nearly flat because they update fixed-size state. Prefill is less monotonic at short lengths because launch overhead and kernel occupancy dominate before the matrix operations reach their efficient regime. These timings use batch size 1, 32 generated tokens, and one recorded timing sample per cell. They are descriptive measurements, not confidence intervals.

Table 14: Detailed prefill and decode scaling from 2K to 128K tokens on one NVIDIA L40S GPU (batch size 1, 32 generated tokens).

Model	Metric	2K	4K	8K	16K	32K	64K	128K
Polar	Prefill latency (s)	0.131	0.031	0.051	0.101	0.248	0.601	1.619
	Prefill throughput (tok/s)	15,640	133,884	160,145	161,551	132,091	109,050	80,959
	Decode latency (ms/tok)	2.27	2.49	2.88	3.70	5.66	9.22	16.30
	Decode throughput (tok/s)	441.2	402.3	346.6	270.1	176.8	108.5	61.3
NoPE	Prefill latency (s)	0.157	0.028	0.049	0.097	0.230	0.576	1.473
	Prefill throughput (tok/s)	13,070	146,568	166,504	168,510	142,319	113,810	89,013
	Decode latency (ms/tok)	2.22	2.45	2.83	3.68	5.62	9.22	16.40
	Decode throughput (tok/s)	450.1	408.8	353.0	271.7	177.8	108.4	61.0
RoPE	Prefill latency (s)	0.160	0.028	0.049	0.096	0.232	0.566	1.472
	Prefill throughput (tok/s)	12,761	147,089	168,287	170,936	141,249	115,786	89,015
	Decode latency (ms/tok)	2.34	2.53	2.95	3.79	5.73	9.33	16.51
	Decode throughput (tok/s)	427.4	395.3	338.7	263.7	174.5	107.2	60.6
Atma-Raven-Titans	Prefill latency (s)	0.198	0.041	0.059	0.111	0.247	0.499	0.999
	Prefill throughput (tok/s)	10,334	100,989	137,903	147,434	132,916	131,464	131,195
	Decode latency (ms/tok)	2.24	2.25	2.24	2.24	2.26	2.25	2.27
	Decode throughput (tok/s)	446.6	445.1	446.0	447.1	442.6	444.8	440.3
Raven Native	Prefill latency (s)	0.214	0.054	0.070	0.139	0.320	0.672	1.357
	Prefill throughput (tok/s)	9,558	75,738	116,378	117,605	102,354	97,527	96,559
	Decode latency (ms/tok)	3.26	3.25	3.24	3.24	3.25	3.25	3.27
	Decode throughput (tok/s)	306.4	307.5	308.2	308.3	307.6	307.5	306.1

H.2 Production-framework stress comparison

To test whether the custom kernels introduce an obvious systems bottleneck at a larger shape, we configured ATMA’s inference engine with a 9.2B-parameter stress shape and compared it with vLLM 0.25.0 serving Qwen3.5-9B on the same L40S in BF16 (Kwon et al., 2023). The models are not weight- or architecture-matched, so Table 15 is a backend stress comparison rather than a model-quality or universal serving claim. ATMA trails on short homogeneous prefill, is similar on long heterogeneous prefill, and leads the recorded high-batch decode cells.

Phase	Batch/workload	ATMA engine	vLLM/Qwen3.5-9B	Ratio
Prefill	Homogeneous $B = 8, T = 512$ (4,096 tok)	11,947 tok/s	12,772 tok/s	0.94×
	Heterogeneous short-heavy (816 tok)	5,752 tok/s	8,002 tok/s	0.72×
	Heterogeneous mixed (4,384 tok)	12,779 tok/s	11,996 tok/s	1.07×
	Heterogeneous long-tail (8,192 tok)	12,251 tok/s	11,721 tok/s	1.05×
Decode, context 512	Batch 128	2,939 tok/s	2,247 tok/s	1.31×
	Batch 256	4,501 tok/s	2,921 tok/s	1.54×
	Batch 512	5,919 tok/s	2,558 tok/s	2.31×
	Heterogeneous $B = 128$ (context 64–1536)	2,950 tok/s	2,403 tok/s	1.23×

Table 15: Recorded systems-scale throughput for ATMA’s 9.2B stress shape and vLLM 0.25.0 serving Qwen3.5-9B on one L40S (BF16, greedy sampling). Ratio is ATMA/vLLM.

I Stress and Environment Diagnostics

FEA Structural Stress Probing: Local Block Secant Gains G Across Length (2K -> 256K)

Polar (ATMA): Flat Gains Across Length									RoPE: Representational Attenuation									NoPE (L40S mbs16): Block 7 Yield Locus								
	2K	4K	8K	16K	32K	64K	128K	256K		2K	4K	8K	16K	32K	64K	128K	256K		2K	4K	8K	16K	32K	64K	128K	256K
B0	0.86	0.86	0.86	0.86	0.86	0.86	0.86	0.86	B0	0.94	0.94	0.94	0.94	0.94	0.94	0.94	0.94	B0	0.91	0.91	0.91	0.91	0.91	0.91	0.91	0.91
B1	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	B1	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	B1	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88
B2	0.96	0.95	0.95	0.95	0.96	0.96	0.96	0.96	B2	1.01	1.00	0.99	0.98	0.98	0.98	0.98	0.98	B2	1.02	1.02	1.02	1.02	1.02	1.03	1.03	1.03
B3	0.90	0.89	0.89	0.89	0.88	0.88	0.88	0.87	B3	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	B3	0.98	0.98	0.98	0.97	0.97	0.97	0.97	0.97
B4	0.87	0.87	0.86	0.86	0.86	0.85	0.85	0.85	B4	1.05	1.07	1.08	1.07	1.06	1.04	1.03	1.03	B4	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.95
B5	0.86	0.85	0.85	0.84	0.84	0.84	0.84	0.84	B5	0.94	0.93	0.93	0.93	0.93	0.93	0.93	0.92	B5	0.94	0.94	0.94	0.94	0.95	0.96	0.96	0.96
B6	1.02	1.02	1.03	1.05	1.07	1.09	1.10	1.11	B6	1.09	1.06	1.04	1.02	1.01	1.01	1.00	1.00	B6	1.13	1.11	1.09	1.09	1.08	1.09	1.09	1.10
B7	0.98	0.97	0.97	0.97	0.97	0.97	0.97	0.97	B7	1.01	1.01	1.00	1.00	1.00	1.00	0.99	0.99	B7	1.02	1.02	1.02	1.03	1.09	1.20	1.31	1.38
B8	0.96	0.96	0.95	0.95	0.95	0.95	0.95	0.94	B8	1.01	1.01	1.00	1.00	0.99	0.99	0.99	0.99	B8	1.01	1.01	1.01	1.00	1.00	1.00	1.00	0.99
B9	0.93	0.92	0.92	0.92	0.92	0.92	0.92	0.91	B9	0.98	0.98	0.98	0.97	0.97	0.97	0.97	0.96	B9	0.99	0.99	0.99	0.98	0.98	0.98	0.98	0.99
B10	1.10	1.10	1.10	1.10	1.10	1.10	1.10	1.09	B10	1.09	1.08	1.07	1.06	1.06	1.05	1.04	1.04	B10	1.16	1.14	1.12	1.12	1.12	1.11	1.11	1.11
B11	1.02	1.02	1.02	1.02	1.02	1.01	1.01	1.01	B11	1.02	1.02	1.02	1.01	1.01	1.01	1.00	1.00	B11	1.04	1.03	1.03	1.02	1.01	1.01	1.00	1.00
B12	1.02	1.01	1.01	1.01	1.01	1.00	1.00	1.00	B12	1.02	1.02	1.01	1.01	1.01	1.00	1.00	1.00	B12	1.04	1.04	1.03	1.02	1.01	1.01	1.01	1.00
B13	1.01	1.01	1.00	1.00	1.00	1.00	1.00	1.00	B13	1.02	1.01	1.01	1.00	1.00	1.00	1.00	1.00	B13	1.04	1.04	1.06	1.08	1.09	1.07	1.06	1.06
B14	1.43	1.44	1.45	1.44	1.41	1.39	1.38	1.37	B14	1.32	1.31	1.29	1.28	1.27	1.26	1.25	1.25	B14	1.39	1.37	1.35	1.35	1.30	1.26	1.24	1.23
B15	1.23	1.23	1.22	1.21	1.20	1.19	1.18	1.17	B15	1.21	1.20	1.19	1.19	1.18	1.17	1.17	1.16	B15	1.18	1.18	1.17	1.14	1.12	1.11	1.09	1.08

Figure 8: Block-wise sampled secant gains over 64 nested FinePDFs documents and 64 perturbation directions per document. The probe localizes checkpoint-specific changes but does not estimate a global Lipschitz constant.

Figure 8 reports the sampled gains. For each block f_i , the diagnostic samples

$$G_i = \frac{\|f_i(x + \delta x) - f_i(x)\|_{\text{rms}}}{\|\delta x\|_{\text{rms}}}, \quad \|\delta x\|_{\text{rms}} = 0.02\|x\|_{\text{rms}}. \quad (10)$$

The maximum sampled gain is a lower bound on the local operator norm along sampled directions. It is not a tight bound on the block Lipschitz constant. In the observed NoPE collapse, changes first concentrate near blocks 6–7 and later propagate through the residual stream. Polar’s sampled gains and activation moments remain comparatively flat through 256K.

J Checkpoint-Variability Diagnostic Journey

The variability finding emerged sequentially. Stage I used L4 GPUs, followed by the first 10B NoPE control on an L4. The anomaly became visible only after the remaining training moved to L40S: short-context convergence remained nearly unchanged while the 256K extrapolation results diverged. Table 16 records the subsequent elimination sequence. The recorded run configurations do not provide a paired controlled seed, so random initialization and the resulting stochastic trajectory remain confounded with the infrastructure transition.

Step	Observation or intervention	Diagnostic implication
L4 baseline	Stage I and the first 10B NoPE control used L4, microbatch 4; 256K NLL was 4.07.	Established the original extrapolation behavior, but not a replicated device-class estimate.
L40S transition	The promoted NoPE run used L40S, microbatch 16; 2K validation ended at 2.8126 versus 2.8160 on L4, but 256K NLL rose to 10.77.	Revealed that nearly coincident short-context curves did not predict the long-context endpoint.
Evaluation replay	L4 and L40S checkpoints were re-evaluated under PyTorch 2.12 and 2.13 with unchanged scores.	Excluded evaluation-version mismatch for these checkpoints.
Microbatch control	NoPE was retrained on L40S with microbatch 4; 2K validation ended at 2.8128 and 256K NLL at 9.90.	Excluded microbatch size alone as the explanation in this comparison.
Deterministic fixtures	Synthetic 2K stream fixtures across the tested versions and microbatches showed no gradient or loss divergence.	Found no short-context execution discrepancy; did not test the full stochastic training trajectory.

Table 16: Diagnostic journey from the L4 control to the L40S follow-ups. Each implication is limited to the intervention actually performed.

All five headline models were trained in the same L40S software and hardware environment, so their primary comparison is internally matched. The earlier L4 checkpoint is retained only as a post-hoc case study. It motivates a broader audit principle: long-context extrapolation claims should report device class, kernels, seeds, and extended-context diagnostics rather than infer robustness from short-context loss alone. The present evidence does not identify CUDA reduction order, floating-point accumulation, device class, or any other single mechanism as causal because the L4/L40S runs were neither initialized from a recorded common state nor replicated across seeds and kernel configurations.